

BIRD CALL CLASSIFICATION USING CONVOLUTIONAL NEURAL NETWORKS

Abstract

Convolutional neural networks (CNNs) are used to classify bird calls from 12 bird species commonly found in the Seattle area. The audio recordings are represented as mel spectrograms, which allow the model to learn frequency and timing patterns from the sounds.

The project includes both a binary classification task and a multiclass classification task. First, a model is trained to distinguish between two bird species. Then, the approach is expanded to classify all 12 species. Finally, the trained model is tested on three unlabeled audio recordings to evaluate how well it performs on unseen data.

The binary classifier achieved a validation accuracy of 85% distinguishing American Crow from American Robin. The 12-class multiclass model achieved 55% validation accuracy on the imbalanced dataset, with species performance ranging from 14% recall for the Northern Flicker to 73% for the House Sparrow.

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1.0 INTRODUCTION

Bird species identification from audio is an important problem in ecology and wildlife monitoring. Traditionally, this task has been performed manually by expert ornithologists who listen to bird calls and songs to identify species. However, manual identification is time consuming, requires specialized expertise, and is difficult to scale for large audio datasets collected in the field. Automated classification using machine learning provides a scalable alternative that can rapidly process large amounts of audio data while maintaining high accuracy.

The dataset used in this project comes from Xeno-Canto [2], a crowd-sourced archive of bird sound recordings collected from contributors around the world. The sound clip is subsampled to 22050 Hz and dataset contains an uneven number of samples per species ranging from 37 to 630, resulting in 1,981 total samples. Audio recordings were converted into mel spectrograms to prepare them for machine learning analysis. Spectrograms represent sound as a two-dimensional image showing frequency changes over time, making them well suited for image-based deep learning methods such as convolutional neural networks.

This project involved building multiple deep learning models for bird sound classification. First, a binary classifier was created to distinguish between two bird species: American crow and American Robin. Next, a larger 12-class classifier was developed to identify all species included in the dataset. Multiple convolutional neural network architectures and hyperparameter settings were compared to evaluate performance improvements. The models were implemented using PyTorch, a deep learning framework that provides fine-grained control over model architecture and training. Finally, the trained models were applied to three unlabeled mystery recordings in order to predict and analyze the bird species present in each recording.

2.0 THEORETICAL BACKGROUND

2.1 Spectrograms and the Mel Scale

A spectrogram is a visual representation of sound showing how frequencies change over time, with the horizontal axis representing time, the vertical axis representing frequency, and color intensity representing sound strength. In this project, bird vocalizations were converted into mel spectrograms using the mel scale, which better matches human auditory perception by emphasizing lower frequencies. Each spectrogram was standardized to 517-time steps by 128 mel frequency bins, representing a 3-second audio window sampled at 22,050 Hz, allowing spectrograms to be treated as images suitable for convolutional neural networks.

2.2 Neural Networks and Training

A neural network consists of interconnected neurons, each of which multiplies inputs by learned weights, adds a bias, and passes the result through an activation function. A single neuron computes a weighted sum of its inputs plus a bias term, expressed as $z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$, and passes this through a nonlinear activation function $f(z)$ to produce an output. This nonlinearity is essential without it, stacking multiple layers would be equivalent to a single linear transformation and the network could not learn complex patterns. The Rectified Linear Unit (ReLU) is commonly used because it reduces vanishing gradient problems and enables efficient learning of complex patterns. As described in Chapter 10 [1, Section 10.7], networks learn by adjusting weights to

minimize a loss function through backpropagation, which calculates how much each weight contributed to the prediction error and updates them using gradient descent. The learning rate controls the size of each update step — too high causes instability and overshooting, too low causes slow convergence. In this project, a learning rate of 0.01 caused complete training failure, confirming its critical role in optimization (James et al., 2021, Section 10.7)

2.3 Convolutional Neural Networks

A convolutional neural network consists of an input layer, convolutional layers, pooling layers, and fully connected output layers. Convolutional layers apply small learnable filters across the input to detect local spatial patterns such as edges and frequency structures. Pooling layers reduce spatial dimensions while retaining the most important features. Fully connected layers then map these extracted features to final class predictions. Additional components can improve training — Batch Normalization stabilizes learning by normalizing layer outputs, and Dropout prevents overfitting by randomly deactivating neurons during training. A dropout rate of 0.5 was chosen as it is a widely recommended default that balances regularization strength with model capacity [1, Section 10.7]. CNNs are particularly well suited to spectrogram data because bird calls contain meaningful local patterns in both time and frequency that convolutional filters can learn automatically [1, Section 10.3].

2.4 Loss Functions and Evaluation Metrics

A loss function measures prediction error and guides training. Binary Cross Entropy (BCE) is used for binary classification, comparing a single sigmoid probability to a binary label. Cross Entropy loss is used for multiclass classification, measuring confidence in the correct class across all outputs. Model performance is evaluated using precision, recall, and F1 score — precision measures correct positive predictions, recall measures identified actual positives, and F1 combines both as their harmonic mean. A confusion matrix visualizes correct and incorrect predictions across all classes, providing more insight than accuracy alone.

3.0 METHODOLOGY

3.1 Data Source and Description

The dataset was sourced from Xeno-Canto via the Kaggle birdcall competition and provided in HDF5 format. Each spectrogram has 128 frequency bins and 517-time steps, representing a 3-second audio window subsampled at 22,050 Hz. The number of samples per species is uneven, ranging from 37 for the Northern Flicker to 630 for the House Sparrow, for a total of 1,981 samples across 12 species [2]

3.2 Preprocessing

Although the spectrograms were precomputed, the time dimension varied across species while the frequency dimension remained fixed at 128 bins for all species. Each spectrogram was stored as a (128 frequency, time) array where the time dimension ranged from 37 to 630 steps depending on the species. To standardize the input, all spectrograms were resized to a uniform shape of (128, 517) using the PIL library, matching the 128 frequency bins and 517 time steps described in the assignment preprocessing pipeline. Values were then normalized to a range between 0 and 1 to improve numerical stability during training. A channel dimension was added to match the required

PyTorch CNN input format of (batch, channels, height, width). The dataset was split into training and test sets using an 80/20 stratified split, producing 1584 training samples and 397 test samples while preserving the proportional class distribution across all 12 species. PyTorch DataLoaders were created with a batch size of 32, with shuffling enabled for training and disabled for evaluation.

3.3 Binary Model:

For the binary classification task, two species—American Crow and American Robin—were selected because they are acoustically distinct and show visually different frequency patterns in their spectrograms, making them a strong and interpretable baseline for the binary task. The architecture consisted of three convolutional layers with ReLU activation, followed by MaxPooling, Adaptive Average Pooling, a Dropout layer with a rate of 0.5, and a sigmoid output layer for binary prediction. The model was trained using BCELoss with the Adam optimizer and a learning rate of 0.001 for 20 epochs. Three binary CNN models were compared: the original model with a learning rate of 0.001 and three convolutional layers, a high-learning-rate version with a high learning rate of 0.01 to examine the effects of faster but potentially unstable training, and a simplified model with only two convolutional layers to figure out how reducing model complexity affects performance.

3.4 Multiclass Model

The multiclass model extended the task to classify all 12 bird species simultaneously. It used the same base convolutional architecture, with the addition of Batch Normalization layers and a fully connected layer with 256 neurons before the final output layer of 12 neurons. This model used CrossEntropyLoss and the Adam optimizer with a learning rate of 0.001. A learning rate scheduler was also applied, reducing the learning rate by half every 10 epochs to improve convergence. The model was trained for 30 epochs, and three different configurations were tested to compare performance under varying training conditions and architectural choices. The three configurations tested were the original model, a deeper four-layer network, and a lower learning rate variant of 0.0001.

3.5 — Test MP3 Processing

To evaluate real-world performance, three unlabeled MP3 audio files were provided as test inputs. These files were loaded using librosa at a sample rate of 22050 Hz. A sliding window approach was applied using 3-second segments with a 1-second stride to capture multiple overlapping audio frames. Each segment was converted into a mel spectrogram, resized to match the training input shape, and normalized. The processed segments were passed through the trained 12-class model, and final predictions were determined using a majority vote across all windows. Additionally, probability distributions were plotted to analyze prediction confidence and assess model reliability across different audio segments.

4.0 RESULTS

4.1 Spectrograms overview

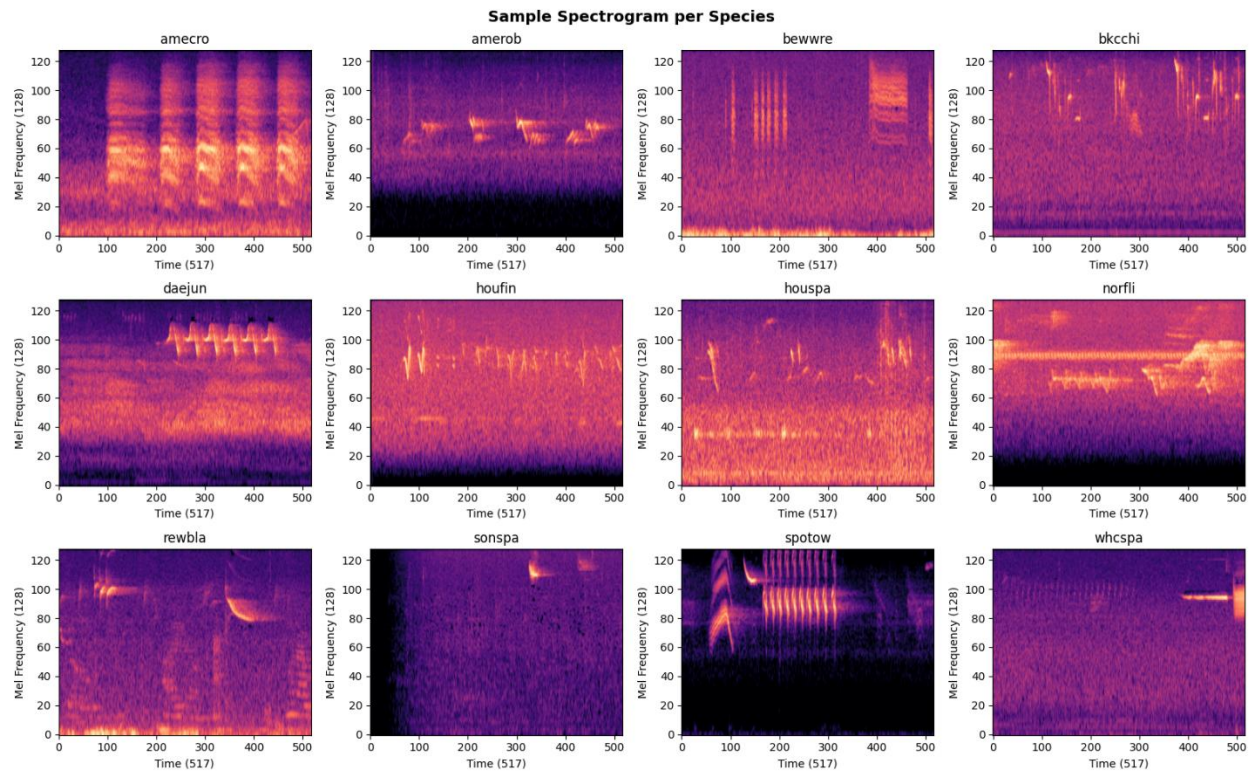


Figure 1: Sample Spectrogram for all bird species

Figure 1 shows a correctly oriented mel spectrogram for each species with frequency on the Y axis and time on the X axis. The varying time axis length reflects the different number of samples per species. Clear differences in frequency distribution and pattern density are visible across species.

4.2 Binary Training Curves

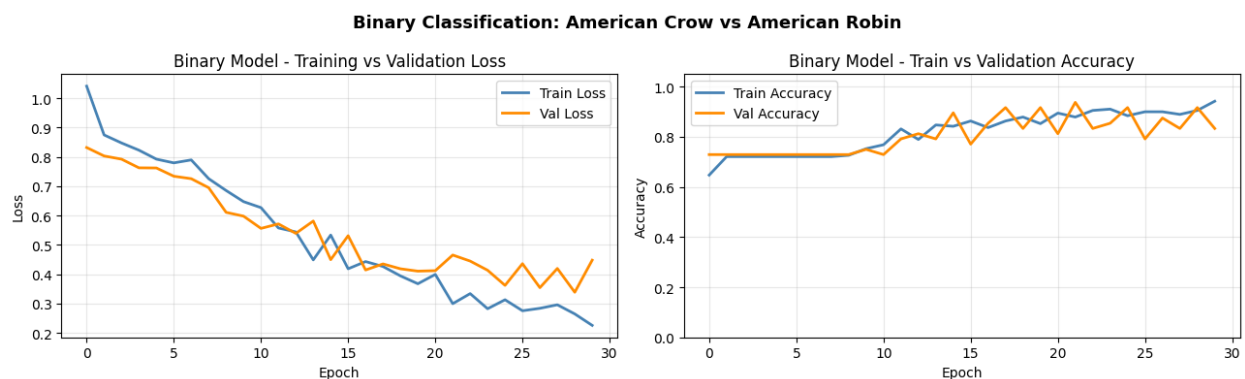


Figure 2: Binary Model- Training/Validation Loss and Test/Validation Accuracy

Figure 2 illustrates the training and validation performance of the model over 30 epochs. Both the training and validation loss decrease consistently throughout training, and the two curves remain relatively close, suggesting that the model generalizes well and does not experience substantial overfitting. Similarly, training and validation accuracy increase steadily from about 70% at the start

of training to approximately 85–90% by the final epochs. While a small gap between training and validation accuracy emerges near epoch 25, indicating the onset of mild overfitting, the difference remains limited

4.3 Binary Comparison

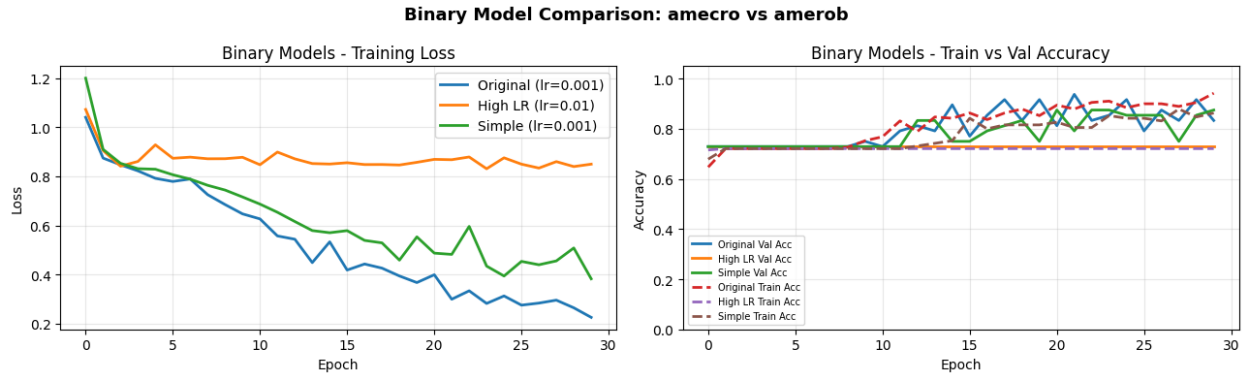


Figure 3: Binary Model- Comparison Plot

Figure 3 compares three binary CNN models with different learning rates and network structures. The model with a high learning rate (0.01) did not learn well, as its loss stayed around 0.85 and its validation accuracy remained close to 72% throughout training. This suggests that the learning rate was too high for effective training. In contrast, both the original model and the simpler model showed steady improvement, with loss decreasing to below 0.4 by the end of training. The original three-layer model achieved the highest validation accuracy of about 85%, while the simpler two-layer model reached a similar accuracy but showed more variation. These results indicate that the original architecture with a learning rate of 0.001 provides the best performance for this binary classification task.

4.4 Multiclass Comparison

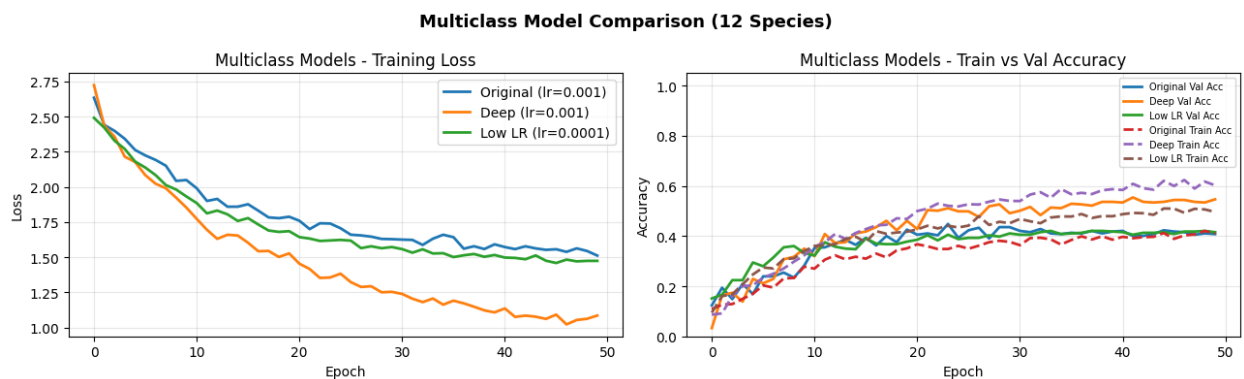


Figure 4: Multiclass Model Comparison- Training Loss and Test Accuracy

Figure 4 compares the performance of three multiclass CNN models trained for 50 epochs. The deeper four-layer model performed the best, achieving the lowest training loss and the highest

validation accuracy of about 55%. In comparison, the original three-layer model and the low learning rate model both leveled off at around 40–45% validation accuracy. All three models showed a widening gap between training and validation accuracy after about 20 epochs, suggesting that some overfitting occurred as training continued. The deep model reached roughly 60% training accuracy while maintaining about 55% validation accuracy, indicating better generalization than the other models despite its longer training time of 275 seconds. The low learning rate model learned more slowly throughout the training process and did not provide any noticeable improvement over the original model.

4.5 Confusion Matrix

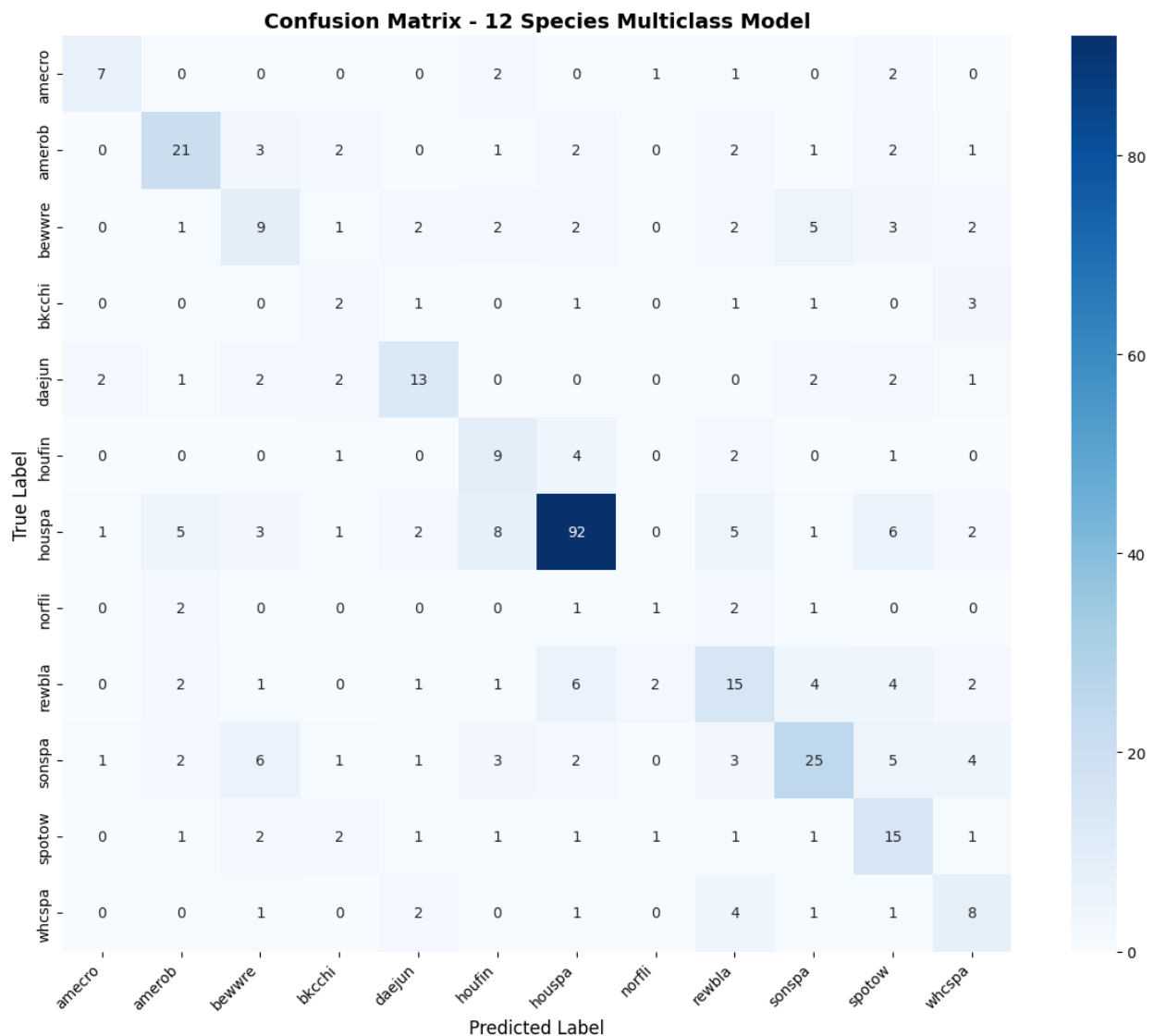


Figure 5: Multiclass Model Confusion Matrix

Figure 5 presents the confusion matrix for the 12-class model tested on 397 images. The values along the diagonal show the number of correct predictions, while the other values show where the

model made mistakes. House Sparrow (houspa) had the best results, with 92 correct predictions out of 126 test images. This species also had the largest training dataset, with 630 samples. On the other hand, Northern Flicker (norfli) and Black-capped Chickadee (bkcchi) had the weakest performance, with only 1 and 2 correct predictions, respectively. These species had the smallest training datasets, containing only 37 and 45 samples. Some species were often confused with one another. For example, Bewick's Wren (bewwre) was frequently predicted as Song Sparrow (sonspa), and Red-winged Blackbird (rewbla) was misclassified as House Sparrow (houspa) several times. Overall, the results suggest that species with more training data were classified more accurately, while species with fewer samples were much harder for the model to identify correctly

4.6 Test predictions

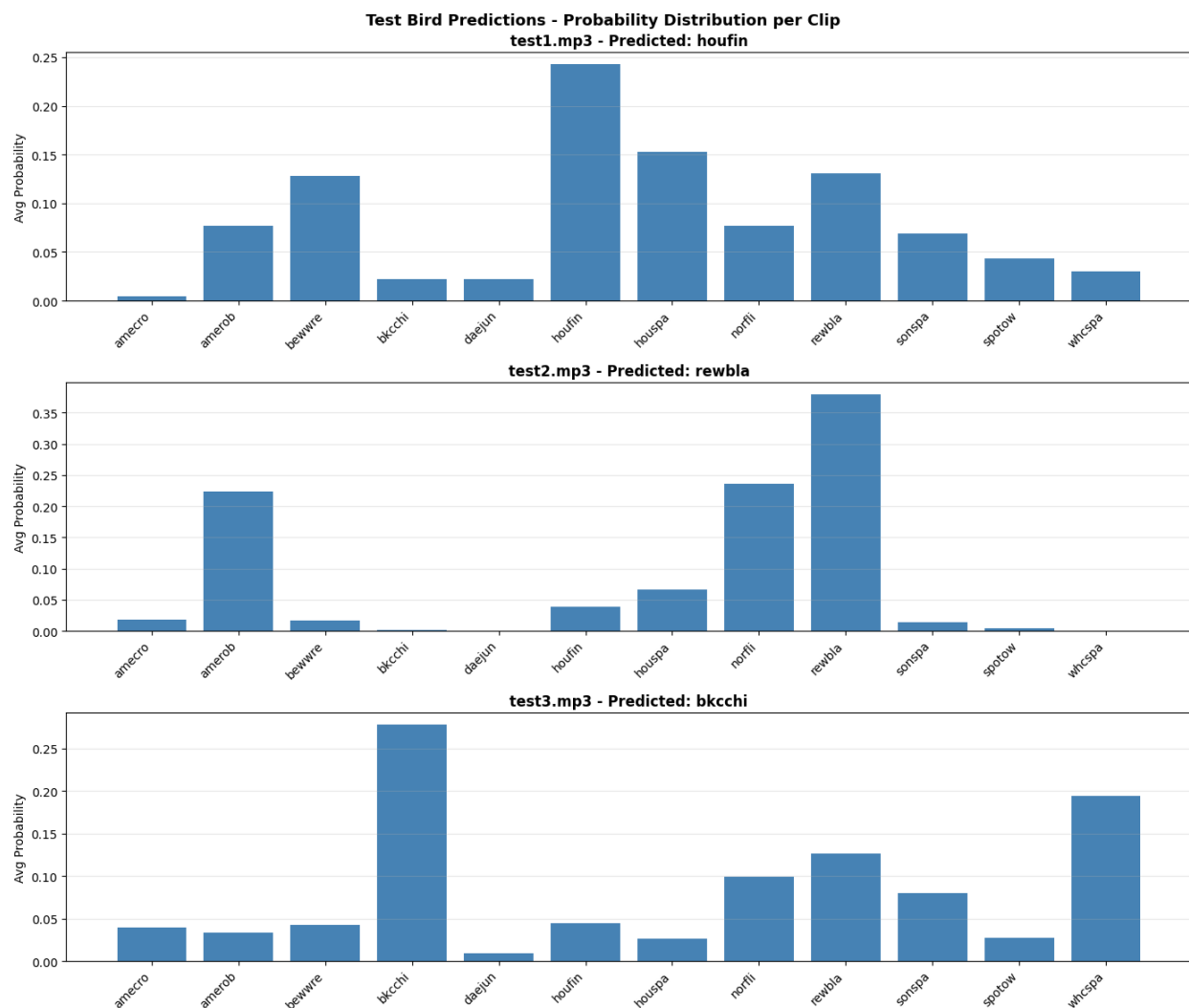


Figure 6: Test Predictions Summary

Figure 6 shows the average prediction results for three test audio clips, based on all their segments. In all cases, the model is not very confident, with the highest probabilities only 24%, 37%, and 27% for test1, test2, and test3. For test1.mp3, the model predicts House Finch (houfin) with 24%, but it also gives noticeable scores to House Sparrow (houspa), Red-winged Blackbird (rewbla), and Bewick's Wren (bewwre), meaning different species may be mixed across its 21 segments. For test2.mp3, the top prediction is Red-winged Blackbird (rewbla) at 37%, but American Robin (amerob) and Northern Flicker (norfli) also get high scores, showing clear uncertainty. For test3.mp3, the model predicts Black-capped Chickadee (bkcchi) at 27%, but White-crowned Sparrow (whcspa) is close behind at about 20%, suggesting these two species may sound similar to the model.

4.7 Test Spectrograms

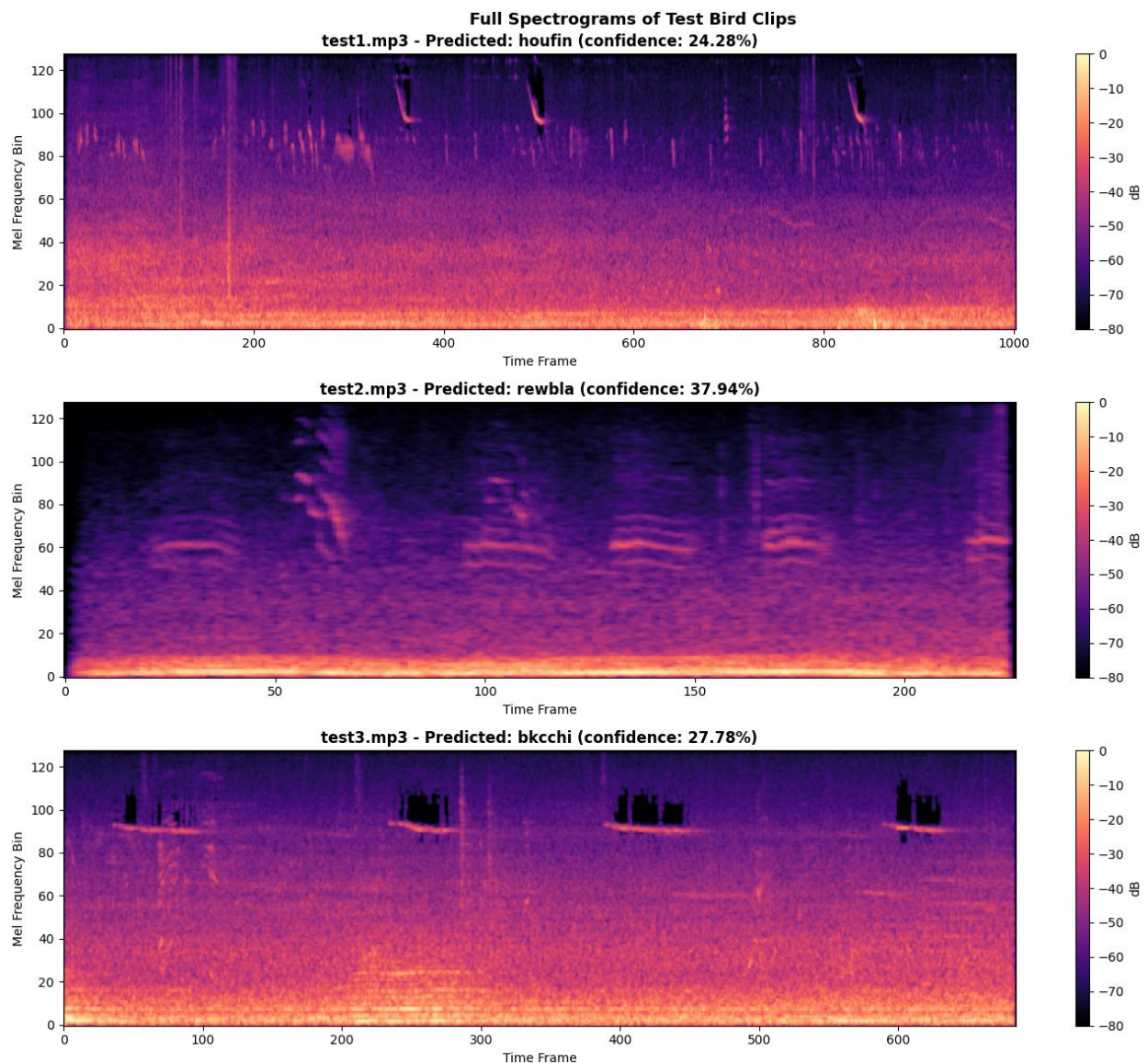


Figure 7 :Spectrograms for the three test recordings used in the final evaluation.

Figure 7 shows the mel spectrograms for the three test audio clips, which basically visualize how sound energy changes over time and frequency. test1.mp3 is the longest clip (about 1000 frames) and shows steady low-frequency sound with several sharp bursts of higher-frequency activity. This suggests there are multiple call events, and possibly more than one bird species in the recording. test2.mp3 is much shorter (around 220 frames) and has a clear repeating curved pattern in the mid-frequency range (about 40–80 mel bins), which looks like a single bird making a consistent type of call. test3.mp3 shows four separate groups of high-frequency activity above mel bin 100, with quiet gaps in between, which looks like short repeated calls from a small songbird. Overall, each recording has a very different pattern, which matches the differences in prediction confidence seen earlier and shows that the clips contain clearly different types of bird sounds.

5.0 DISCUSSION

5.1 Limitations

One key limitation of this project is the relatively limited scope of the dataset and computational training process. Although the dataset contains 12 species with an uneven number of samples ranging from 37 to 630 per class, it is still relatively small compared to real-world bird audio distributions, where species are highly imbalanced and environmental noise is much more variable.

In the binary model, the high learning rate variant of 0.01 produced a loss that remained constant at approximately 0.693, which equals $\ln(2)$. This value corresponds exactly to the loss of a model predicting 50% probability for every sample regardless of input confirming the model learned nothing and was equivalent to random guessing. Additionally, training time constrained experimentation with larger architectures or more extensive hyperparameter tuning. While the models converged quickly and achieved high accuracy, longer training with more varied augmentation strategies could potentially improve robustness and generalization. In terms of computational cost, the binary classification model trained in approximately 12 seconds over 20 epochs, while the multiclass model required approximately 56 seconds over 30 epochs, both using a T4 GPU on Google Colab.

5.2 Species difficulty and confusion

Although overall performance was perfect on the training data, the test clip analysis revealed recurring confusion between the Black-capped Chickadee (bkcchi) and House Sparrow (houspa). Both species produce short, high-pitched vocalizations with overlapping frequency ranges and similar harmonic structures. Listening to their calls on Xeno-Canto confirms these similarities — both are difficult to distinguish by ear in noisy conditions. Among all 12 species, sparrows such as sonspa, whcspa, and houspa are acoustically most similar, sharing comparable call durations and spectral energy distributions. The Northern Flicker and Black-capped Chickadee achieved the lowest recall scores of 14% and 22% respectively, corresponding directly to having the fewest training samples at 37 and 45. The Northern Flicker produces high-pitched repetitive calls — not low frequency as previously stated which are visually distinctive in spectrograms but difficult to learn with so few examples.

5.3 Test Predictions Summary

The low confidence scores across all three clips, ranging from 24% to 38%, reflect the genuine difficulty of the task with an imbalanced training dataset. These results are more cautious and realistic than initially expected, suggesting the model has learned to distribute uncertainty rather than overconfidently predict a single species.

5.4 Multiple birds analysis

The results also highlight the possibility of multiple-bird recordings influencing model uncertainty. In particular, the first test clip showed a more distributed probability spread across several classes and lower confidence in the top prediction. This aligns with the spectrogram analysis, which showed less structured frequency patterns and more variation over time. These characteristics are consistent with overlapping bird calls or background environmental noise. In real-world deployments, such mixed audio conditions would likely reduce classification accuracy unless additional source separation or segmentation techniques are applied.

5.5 Alternative models

Traditional methods such as support vector machines or random forests could be applied to flattened spectrogram features, but these treat each frequency bin independently and miss the spatial relationships that define bird call shapes. Neural networks make particular sense here because bird call classification is fundamentally a pattern recognition problem on 2D spectrogram images. CNNs automatically learn hierarchical spatial features — frequency bursts, harmonics, and temporal patterns — without manual feature engineering, making them the natural choice. Future work could explore ResNet architectures, attention-based models, or SpecAugment[6] data augmentation to improve robustness on noisier recordings.

6.0 CONCLUSION

Overall, this project demonstrates that convolutional neural networks are highly effective for bird species classification using spectrogram representations of audio. The binary classifier achieved 85% validation accuracy distinguishing American Crow from American Robin. The 12-class multiclass model achieved 55% validation accuracy on a heavily imbalanced dataset, with performance ranging from 14% recall for the Northern Flicker to 73% for the House Sparrow. This shows that spectrograms contain strong discriminative features that CNNs can learn efficiently. The comparison of different learning rates and architectures highlighted the importance of stable optimization, with overly high learning rates causing complete training failure and simpler models reducing learning efficiency. The analysis of test MP3 files further showed that the model can generalize to unseen audio segments using a sliding window approach, producing consistent predictions across time. However, cases with lower confidence and more distributed probabilities suggest that real-world audio complexity, including overlapping species and environmental noise, remains a significant challenge. In conclusion, while the model performs exceptionally well on curated data, future improvements in dataset diversity, augmentation, and model architecture would be necessary for deployment in real-world bioacoustic monitoring systems.

7.0 REFERENCES

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